Ogbonna Arinze

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SUMMARY

Senior Fullstack Engineer with 8 years of experience building scalable, AI-powered platforms and enterprise applications. Deep expertise in React, TypeScript, and modern backend architectures (NodeJS, NestJS, PostgreSQL), with a strong foundation in AI/LLM integrations (LangChain, Mastra, RAG pipelines). Adept at driving end-to-end feature delivery, automating complex workflows, and deploying robust infrastructure (Docker, AWS, CI/CD). Highly proficient in AI-augmented development workflows (Cursor, Copilot) and passionate about building products that solve complex regulatory and data challenges.

CORE COMPETENCIES

- **Frontend Development:** React, TypeScript, Next.js, Tailwind CSS, and scalable UI architectures.
- **Backend & APIs:** NodeJS, NestJS, REST APIs, PostgreSQL, and SQL optimization.
- **AI & LLM Systems:** LangChain, Mastra, RAG pipelines, OpenAI API, and Vector Search (pgvector, Chroma).
- **Infrastructure & DevOps:** AWS, Docker, GitHub Actions, CI/CD pipelines, and Nx monorepos.
- **Additional Languages:** Python (Personal Projects), JavaScript (ESNext).
- **Engineering Practices:** AI-augmented workflows (Cursor, Copilot), TDD, Agile, and full product ownership.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Co-developed a full-stack **Next.js** chat widget powered by a Retrieval-Augmented Generation (RAG) system using OpenAI's GPT models, **LangChain**, Vercel AI SDK, and Chroma vector database, enabling users to query complex data in real time.
- Spearheaded the redesign of the main product for connecting and transferring clinical trial sites' data to various upstream EHR systems, migrating from a Vue.js web application to a Chrome extension in **React** and **TypeScript**.
- Architected a reusable design system in React and Radix UI that improved UI consistency, accessibility, and developer efficiency across products.
- Optimized application performance by implementing lazy loading, code splitting, and CDN asset delivery, significantly reducing load times as measured via Lighthouse and Chrome DevTools.
- Increased stakeholder confidence by writing end-to-end tests with Playwright in the CI/CD pipeline, publishing test results on GitHub Pages using the Allure Reports plugin.
- Implemented authentication and authorization workflows on the frontend using JWT-based flows, securely handling access and refresh tokens with HTTP-only cookies.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive user interface using **React**, achieving a 25% boost in user engagement. Collaborated closely with software architects to create a scalable frontend architecture.
- Established engineering standards for code quality, testing, and deployment across the team using ESLint, Prettier, Husky, Lint-staged, and GitHub Actions, reducing errors and accelerating CI/CD processes.
- Optimized frontend performance and load efficiency by identifying and resolving render bottlenecks using React Profiler and Lighthouse Audits, reducing bundle size by 30-40% and achieving a 35% faster load time.
- Enhanced the development workflow by introducing automated testing frameworks, including Jest and Enzyme, which led to a 20% decrease in bug reports.

Founders Lair Beijing, China (Remote)

Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS** RESTful APIs to power new features, focusing on testable, maintainable, and well-documented code.
- Translated business requirements into efficient database models and schemas, collaborating closely with the business team to ensure technical solutions met business needs.
- Transformed user journeys and Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface.
- Engineered a CI/CD pipeline for automated deployment of APIs to testing environments on Heroku and production servers on **AWS**.

ActivEdge Technologies Lagos, Nigeria

Senior React Developer *Oct 2019 - Sep 2021*

- Played a pivotal role in a team of 3 frontend engineers to build a high-volume bank Settlement Solution, significantly enhancing transaction processing capabilities to handle millions of transactions daily, boosting settlement rates by 80%.

- Spearheaded the development of critical modules for the Settlement Solution using **React** with Redux for state management, RxJS for data streaming, and Axios for asynchronous API requests.
- Developed a real-time account reconciliation solution, boosting the reconciliation of daily transactions by 60% through innovative transaction processing and data management strategies.

PROJECTS

OurNigeria ournigeria.ng

Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected and built an AI-powered data analysis platform as an Nx monorepo managing 6 applications, including a **NestJS** API backend, a **Next.js** citizen-facing web app, and a document ingestion pipeline.
- Designed and implemented a multi-agent AI system using the Mastra framework with 5 specialized agents for intent classification and domain-specific RAG-powered analysis.
- Built a hybrid RAG pipeline combining **pgvector** similarity search with BM25 full-text search, merged via Reciprocal Rank Fusion, with Cohere reranking and Voyage AI embeddings.
- Designed the **PostgreSQL** 16 database schema with Prisma v7, including pgvector extension for vector storage, and wrote custom SQL migration tooling.
- Configured CI/CD with GitHub Actions for automated builds and **Docker** Compose production deployment with multi-stage Dockerfiles, health checks, and memory limits.

EDUCATION

University of Nigeria Nsukka, Enugu

BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*